Humans Set Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected permission-and-review audit.

Read the exhibition approval, inspect the room change and complete Jo’s task before considering an unclear result.

Keep one habit: Keep actor, action, version, destination, time and completed uses visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Verify Important Outputs

Generate, Retrieve, or Calculate?

Missing and Unbalanced Examples

W1 · Separate capability, correctness and permission

> Use the supplied exhibition reference.

Your evidence record

- Exhibition reference for W1–W3: checked A says “Model bridge exhibition, Thursday, Room 6.” The teacher responsible for the class board authorises the class editor to post exactly A there today, once, and explicitly excludes online posting. No posting has occurred. A later current room record says Room 8; B changes only that room. The old permission covers exactly A, not B. Keep this case separate from poster P2 and Jo’s library task.

- A tool offers a Public website button. The exhibition facts in A have been verified. The teacher’s approval covers the classroom board only. State what the button, the fact check and the approval each establish.

  Decide whether the editor may publish A publicly.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Read the approval without enlarging it

> Identify every relevant limit and apply permission already given.

Your evidence record

- List the decision maker, allowed actor, version, destination, timing and number of postings in the original exhibition approval.

  Before any posting and with no changes to the facts or **permission**, explain whether the editor can carry out that exact action without asking again.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Review a changed version concretely

> Explain the two different failed checks before requesting review.

Your evidence record

- The room changes to Room 8 and the editor creates B. Explain why publishing old A and publishing unapproved B each fail a different check.

  Write a specific request for the teacher to review B, keeping the original destination and one-posting limit.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the permission check to a new case

> Keep one first attempt before hints or feedback.

Read and explain independently

In a fictional library, the librarian controls the internal shelf list. The librarian tells assistant Jo: “Replace the internal list with checked version L2 on Friday, once.

Do not change the public catalogue.” It is Friday, L2 remains checked, and no replacement has occurred. A tool offers both destinations.

Assess these proposals separately: Jo replaces the internal list with L2; Jo updates the public catalogue with L2; visitor Sam replaces the internal list with L2; Jo replaces the internal list with a changed L3.

Identify the approval detail that settles each case.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the permission check to a new case

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MAIN LESSON · SECTION 16

W5 · Inspect an unclear result after Jo’s task

> Use W4’s once-only permission; check completion before retrying.

Your evidence record

- Jo attempts the permitted internal replacement, but the tool’s response does not show whether it succeeded. Explain why immediately repeating it is a poor choice under a once-only approval. What should Jo check?

  If Jo discovers a public change occurred instead, describe the first recovery steps.

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Complete the sorter's human-control map: **permission**s, review triggers, override, correction, and stop conditions.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Read an approval without enlarging it for Human **authority** and **permission**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Human **authority** and **permission**. Proposed explanation: "A working button grants **permission**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given?

Specify what is needed.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An assistant may suggest object locations; a teacher must approve moving valuable equipment.

What may happen without approval?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorter marks a child's notebook throw away. Who should decide whether it is discarded?
Name a **boundary** the system should enforce.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 28 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given?

Specify what is needed.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ **Reference · Independent-case response copy**

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

In a fictional library, the librarian controls the internal shelf list. The librarian tells assistant Jo: “Replace the internal list with checked version L2 on Friday, once.

Do not change the public catalogue.” It is Friday, L2 remains checked, and no replacement has occurred. A tool offers both destinations.

Assess these proposals separately: Jo replaces the internal list with L2; Jo updates the public catalogue with L2; visitor Sam replaces the internal list with L2; Jo replaces the internal list with a changed L3.

Identify the approval detail that settles each case.

My reasoning and evidence

▼ **Extra practice · Use the worked picture**

REFERENCE · SECTION 18

Extra practice · Use the worked picture

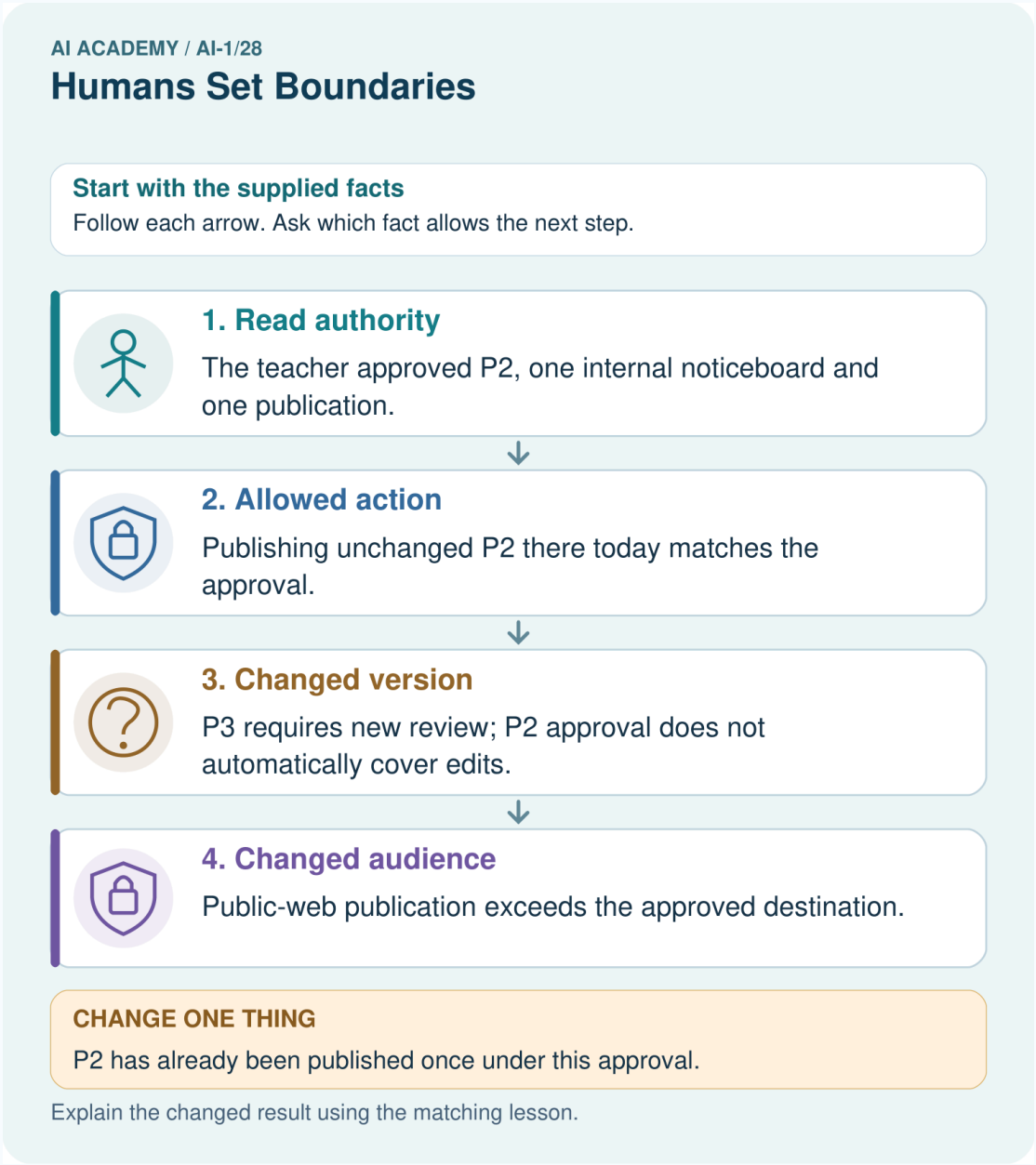
> Keep the explanation and the support you used.

Use this when helpful

A fictional teacher approves one checked classroom poster for internal display.

Approval: publish exactly poster P2 on the classroom noticeboard today, once.

Assistant proposes P2 there; P3 there after an edit; P2 on a public website.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which original proposal is authorised? Why do the other two need separate approval?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Record sorter actions allowed automatically and actions requiring the classroom owner’s explicit review.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Record sorter actions allowed automatically and actions requiring the classroom owner’s explicit review.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional teacher approves one checked classroom poster for internal display.
- Which labelled step matches this detail? Public-web publication exceeds the approved destination.
- 1. Allowed action
- 2. Changed audience
- 3. Read **authority**
- 4. Changed version
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Changed audience Public-web publication exceeds the approved destination. A general intention to help the class does not grant every action that might help.

Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact.

The assistant should hold the changed poster for review rather than treating its own confidence as **permission**.

Changed-case question

- Changed-case practice: P2 has already been published once under this approval. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. A second publication is not covered.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A second publication is not covered. The instruction allowed a single use, not repeated actions.

Repair and save your work

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

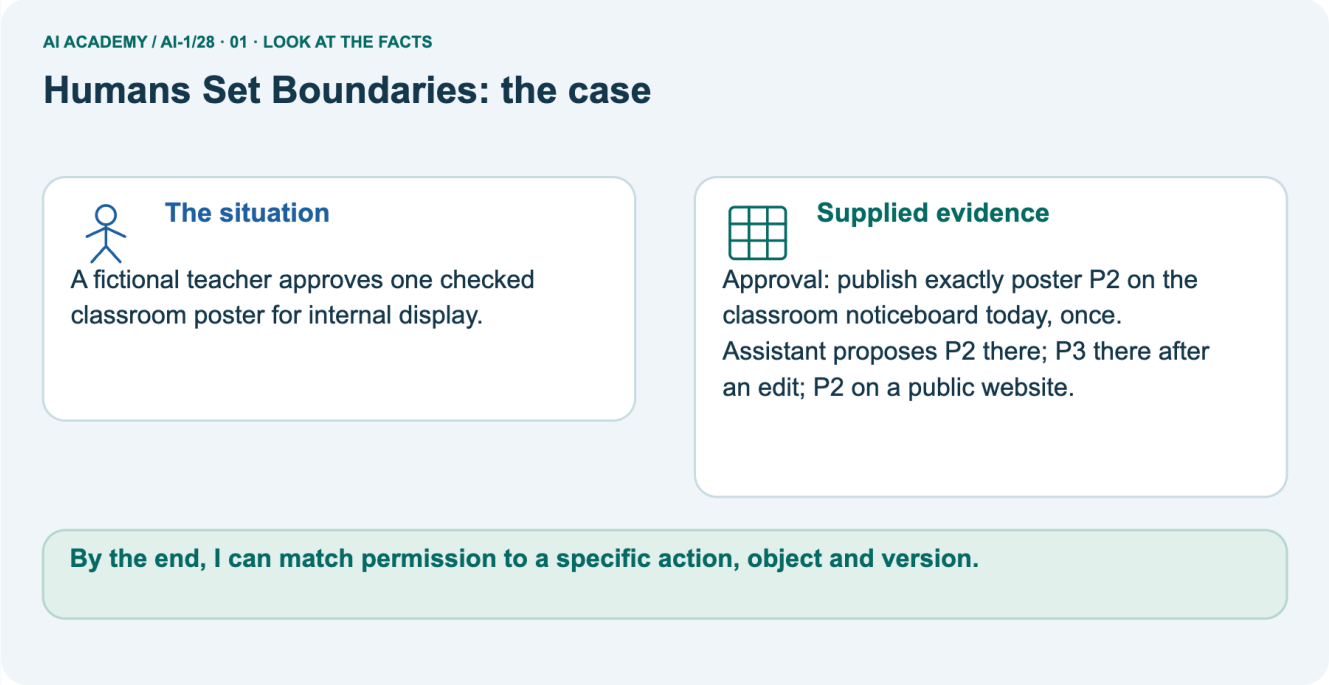
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Humans Set Boundaries: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional teacher approves one checked classroom poster for internal display. Supplied facts: Approval: publish exactly poster P2 on the classroom noticeboard today, once.

Assistant proposes P2 there; P3 there after an edit; P2 on a public website.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **authority**: The teacher approved P2, one internal noticeboard and one publication. → Allowed action: Publishing unchanged P2 there today matches the approval. → Changed version: P3 requires new review; P2 approval does not automatically cover edits. → Changed audience: Public-web publication exceeds the approved destination.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Match the proposed action to its permission

AI ACADEMY / AI-1/28 · 03 · INSPECT THE LESSON MODEL

Match the proposed action to its permission

Proposed action	Check against approval	Decision
Version A on class board	Today, once, by editor	Within scope
Version A on public site	Public posting excluded	Outside scope
Changed version B	Only A was approved	Review changed version
Post A a second time	Once has been used	Outside scope

Each decision uses the approval in this story. An unchanged, unused permission can be acted on without requesting it again.

Follow the labelled picture with the matching explanation. Open the vector diagram

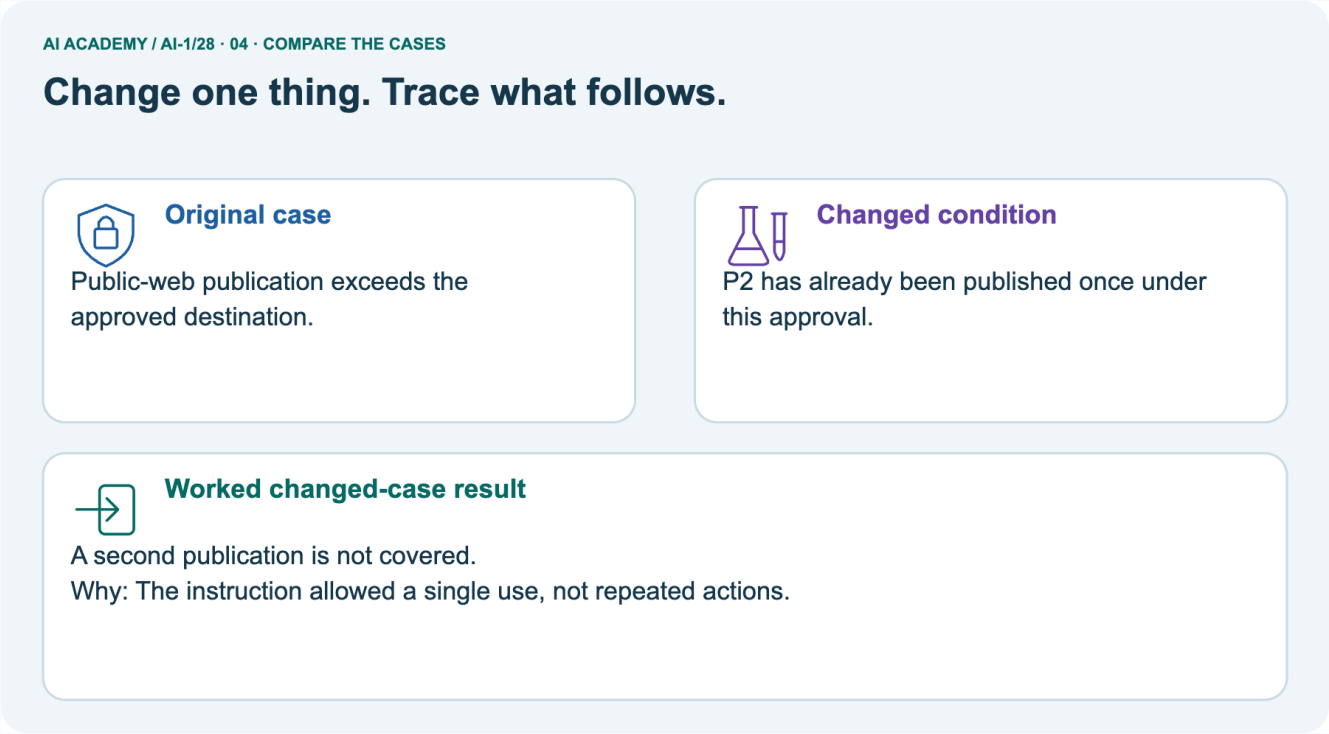
Match the proposed action to its **permission**. Version A on class board | Today, once, by editor | Within **scope**. Version A on public site | Public posting excluded | Outside **scope**.

Changed version B | Only A was approved | Review changed version. Post A a second time | Once has been used | Outside **scope**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

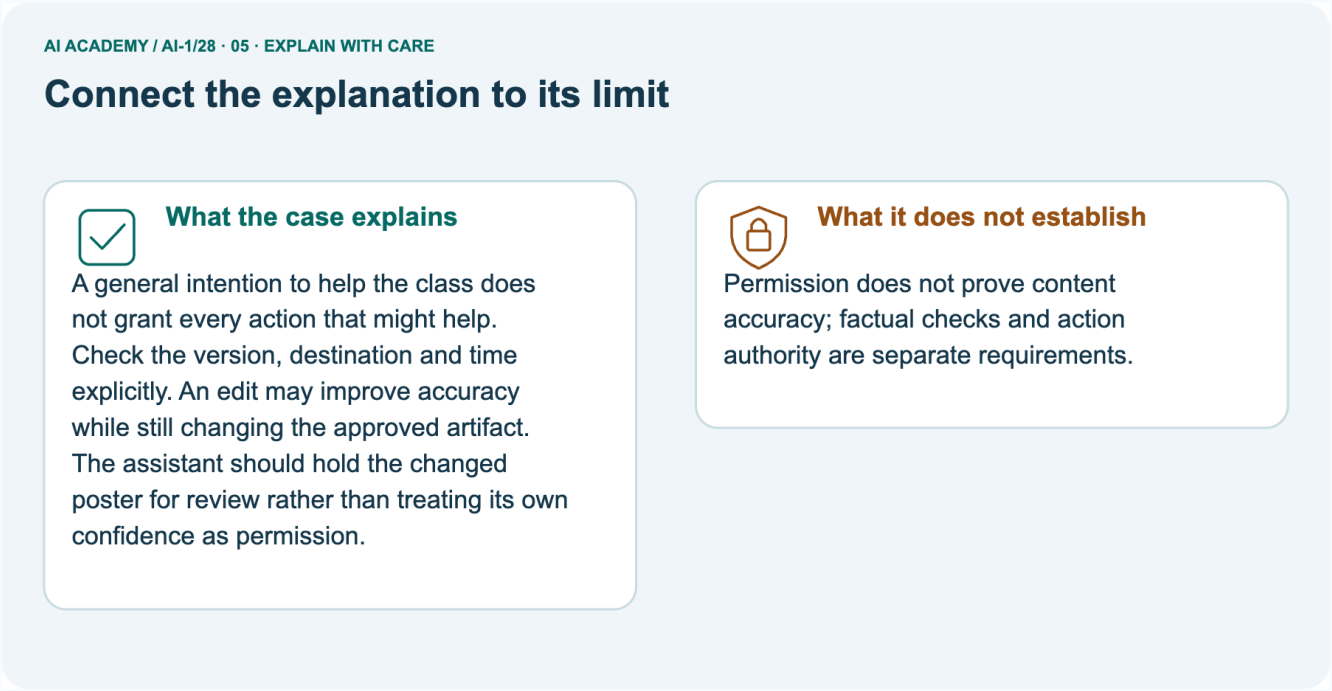
Original case: Public-web publication exceeds the approved destination. Changed condition: P2 has already been published once under this approval. Expected result: A second publication is not covered.

Reason: The instruction allowed a single use, not repeated actions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

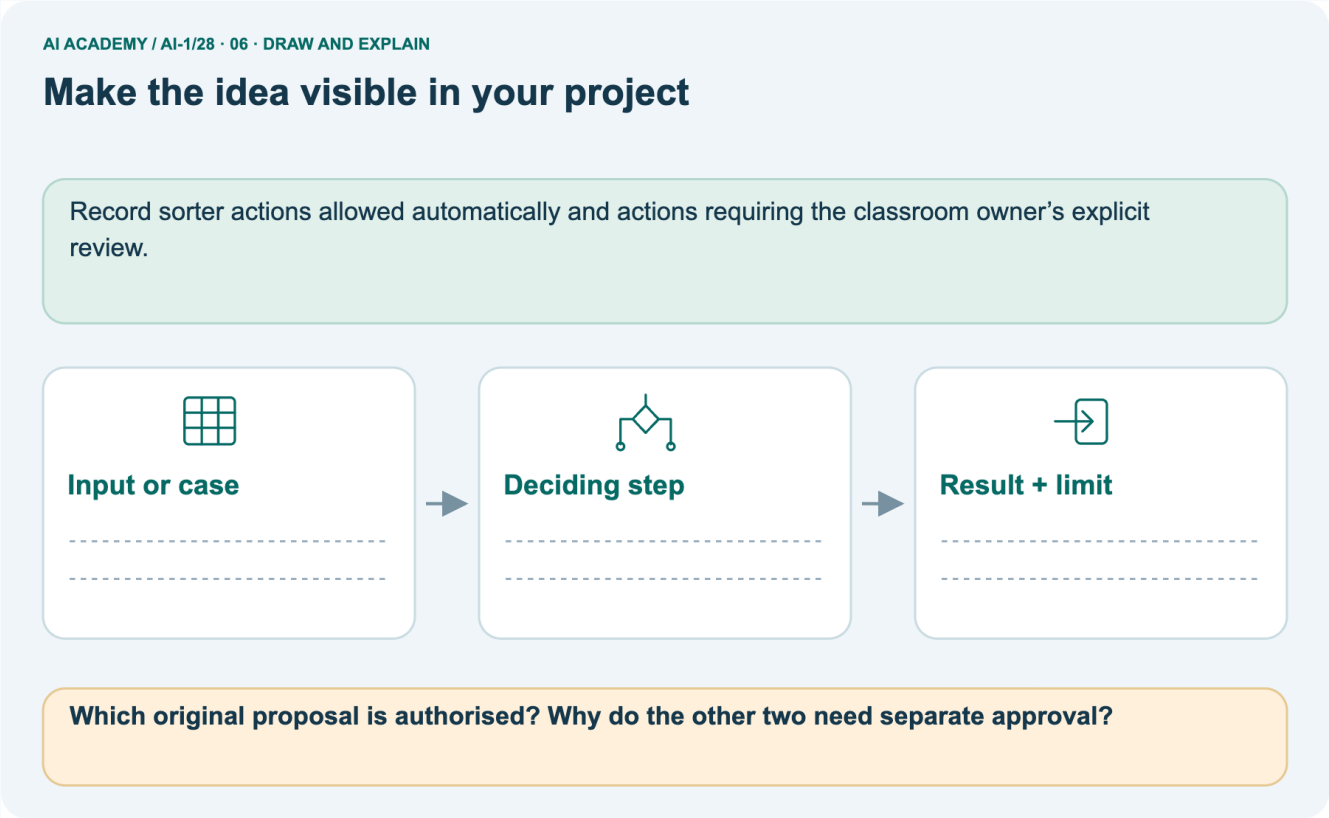
A general intention to help the class does not grant every action that might help. Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact.

The assistant should hold the changed poster for review rather than treating its own confidence as **permission**. Limit: **Permission** does not prove content accuracy; factual checks and action **authority** are separate requirements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Record sorter actions allowed automatically and actions requiring the classroom owner's explicit review. Draw your **input**, deciding step, result and limit. Supported question: Which original proposal is authorised?

Why do the other two need separate approval?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Humans Set Boundaries: the case

02. Follow the deciding steps

03. Match the proposed action to its permission

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.